SULIT

SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

Name	MOHAMED ALIF FATHI BIN ABDUL LATIF
IC No. / Matric No.	041231080417 / A23CS0112
Year / Program	2 / SECPH
Section	01
Lecturer Name	DR. ROZILAWATI BINTI DOLLAH @ MD. ZAIN
Case Study	MODULE LOAN: LOAN APPLICATION AND LOAN APPROVAL
System Name	KADA WEB-BASED COOPERATIVE SYSTEM
Group Name	ALPHATECH

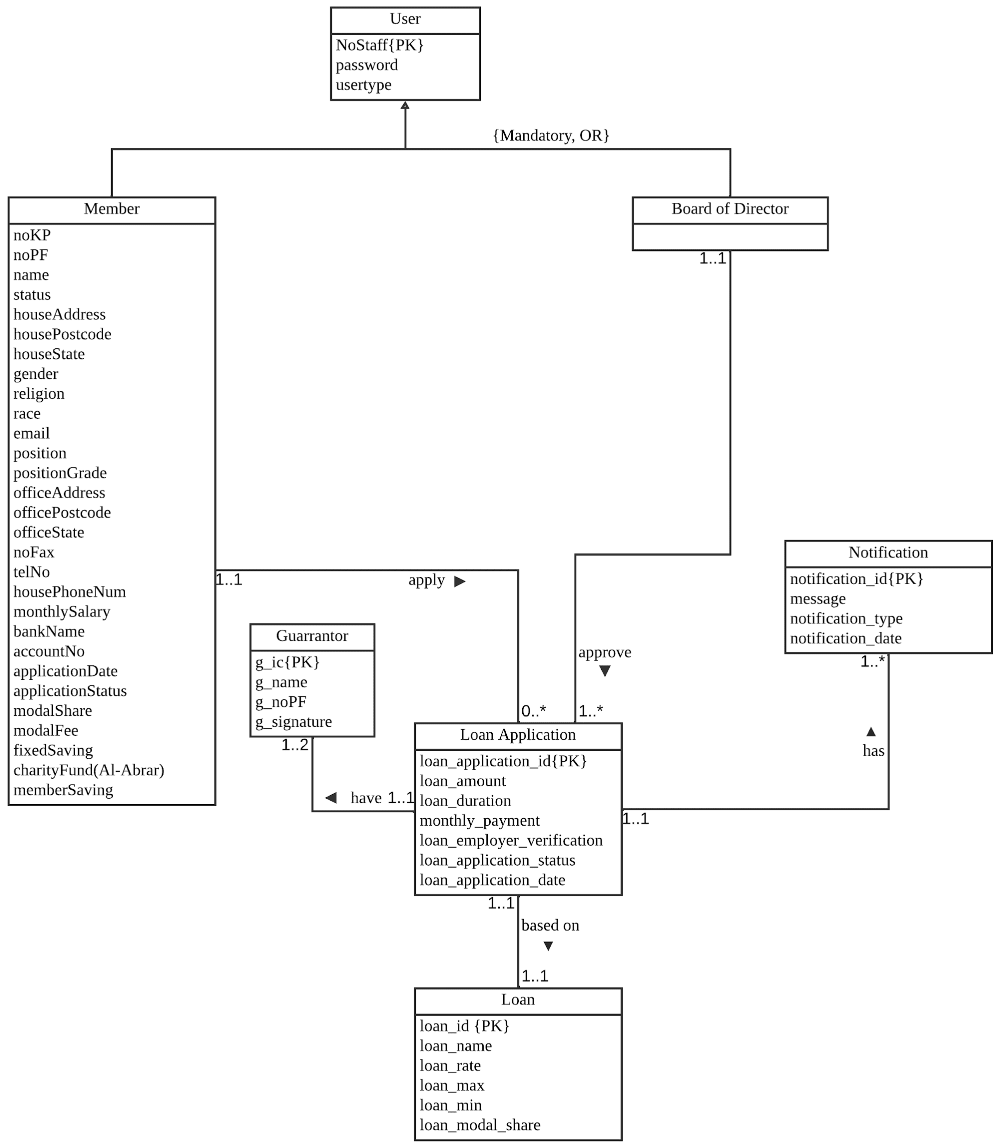
SECTION A (SYSTEM'S OVERVIEW)

1.1	**Overall Description**

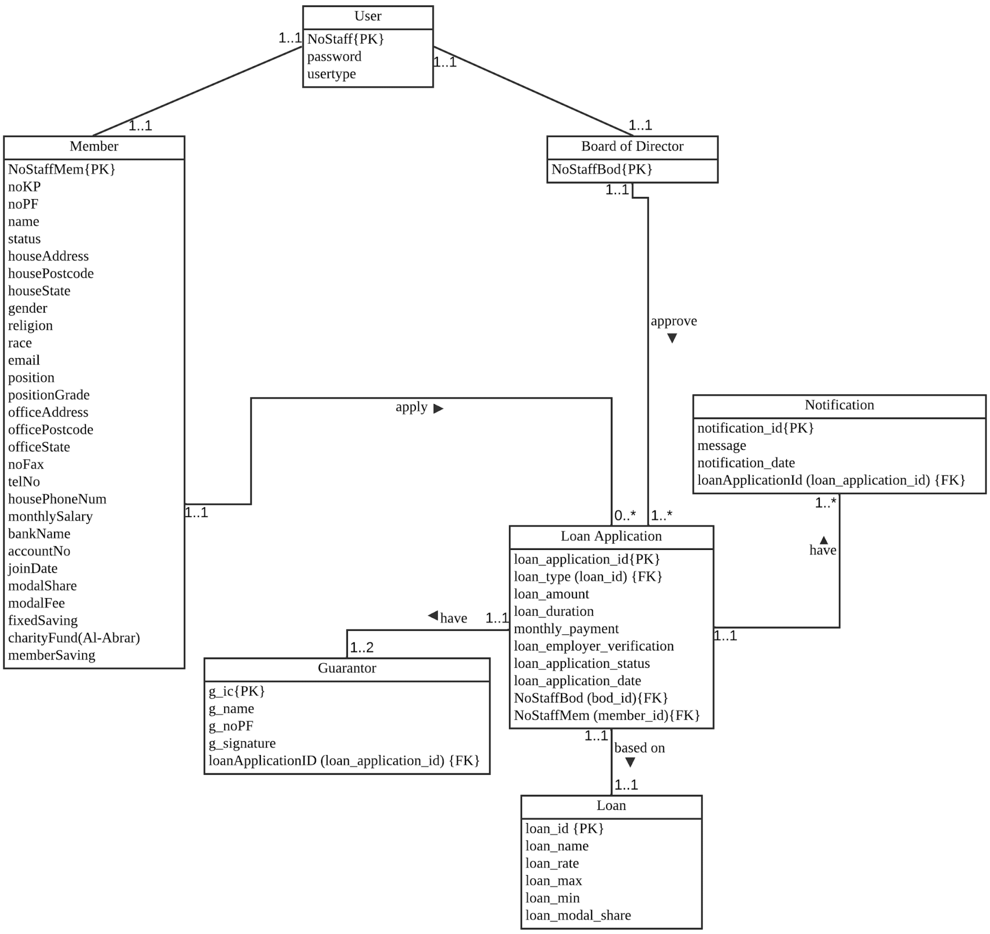
The proposed system is a website designed to digitize KADA's manual processes. The system still maintains some functionality but adds some modification to enhance efficiency. Current issues that prompted the initiation of this project are inconsistent data entry, paper-based system, limited data collection and analysis and high dependence on human resources .The system consists of three users, which are employee, admin and board of directors. For employees, they can apply for membership. After becoming a member, they can apply for loans if they meet the RM300 minimum modal share with RM 50 entry fee. The board of directors can review and approve loan or membership applications online. An admin can manage system policies, view member data such as loan application and membership application, and generate reports of the system. There are four modules for this system which are loan, membership, financial and administration. For module loan, it consists of loan application and loan approval. For module membership, it covers membership application and membership approval. For Financial, it includes generating reports by admin, generating financial statements by members and updating financial details such as loan payments and fee payments. Table below are the details of the group member modules assigned to.

Group Member's Name	Matric No	Module
MOHAMED ALIF FATHI BIN ABDUL LATIF	A23CS0112	Loan
MUHAMMAD AFIQ DANIAL BIN ROZAIDIE	A23CS0117	Membership
IMAN ABADI BIN MOHD NIZWAN	A23CS0084	Financial
AFIF SHAQIR IRFAN BIN ARQAM	A23CS0204	Administration

1.2	**Project Weaknesses or Improvement**
	One of the problems found is members are unaware of the status of their loan applications, which can lead to confusion. This lack reduces member satisfaction and increases the workload for staff as they need to respond to frequent inquiries from members about their loan status. To solve this issue, an automated notification system can be implemented to provide real-time updates via emails for the loan application process. For example, members can be notified when their loan application is received, ongoing, rejected or completed.
1.3	**Proposed Module**
	To solve the current issue in the loan module, a Notification Module is proposed. This addition will provide real-time updates on loan application statuses. The sub-module includes automated notifications through email to inform members of their loan application, such as when the application is received, ongoing, or rejected. A real-time tracking system will be integrated into the member's dashboard. Implementing this sub-module will significantly improve communication and reduce confusion for members. This will keep them informed throughout the loan application process. Additionally, the workload on staff will be reduced by minimizing repetitive inquiries like members need to ask when the loan is approved to be ongoing.
SECTION B (DATABASE PLANNING AND DESIGN)	
2.1	**Conceptual ERD**



2.2	**Logical ERD**



2.3 Data Dictionary for Logical ERD

Table Name	Field Name	Data Type	Description
User	NoStaff {PK}	int	Employees staff ID
	password	varchar(128)	Employees password to login
	usertype	varchar(128)	Employees type (admin, BOD and member)
Member	NoStaffMem{PK}	int	Member staff ID
	noKP	varchar(12)	Member IC number
	noPF	int	Member PF number
	name	varchar(150)	Member name

		status	varchar(20)	Member marital status
		houseAddress	varchar(100)	Member address
		housePostcode	int(5)	Member postcode
		houseState	varchar(100)	Member state
		gender	char(1)	Member gender
		religion	varchar(100)	Member religion
		race	varchar(100)	Member race
		email	varchar(100)	Member email
		position	varchar(100)	Member position in company
		positionGrade	varchar(100)	Member position grade in company
		officeAddress	varchar(100)	Member office address
		officePostcode	int(5)	Member office address postcode
		officeState	varchar(100)	Member office address state
		noFax	int	Member number fax
		telNo	int	Member number phone
		housePhoneNum	int	Member house phone number
		monthlySalary	float	Member salary
		bankName	varchar(100)	Member bank name
		accountNo	int	Member bank account number
		joinDate	timestamp	Member joined date as a membership
		modalShare	float	Member modal share

				in company
		modalFee	float	Member modal fee in company
		fixedSaving	float	Member fixed saving in company
		charityFund(Al-Abrar)	float	Member charity fund in company
		memberSaving	float	Member saving in company
	Board of Director	NoStaffBod{PK}	int	BOD staff ID
	Guarantor	g_ic{PK}	varchar(12)	Guarantor IC number
		g_name	varchar(150)	The name of guarantor
		g_NoPF	int(12)	Guarantor Personal File' number
		g_signature	mediumblob	Guarantor signature
		loanApplicationID {FK}	int	Loan Application ID
	Loan Application	loan_application_ID{PK}	varchar(10)	An id that uniquely identifies loan applications
		loan_type{FK}	int	Loan type ID
		loan_amount	double	Amount requested for a loan
		loan_duration	int	Period of time the loan in scheduled for the loan payment in months
		monthly_payment	float	amount of money required to paid monthly based on the loan amount and duration
		loan_employer_verific	mediumblob	Employer

	ation		verification
	loan_application_status	varchar(30)	Loan status
	loan_application_date	timestamp	date of loan application
	NoStaffBod{FK}	int	BOD staff ID
	NoStaffMem{FK}	int	Member staff ID
Notification	notification_id{PK}	int	notification ID
	message	text	Message to loan applicant
	notification_date	timestamp	Notification date
	loanApplicationId{FK}	int	Loan Application ID
Loan	loan_id {PK}	int	Loan ID
	loan_name	varchar(50)	Loan name
	loan_rate	float	Loan rate
	loan_max	float	Loan maximum amount
	loan_min	float	Loan minimum amount
	loan_modal_share	float	Loan minimum modal share needed

2.4 **Relational Database Schema**

Relation:

User(NoStaff , password, usertype)
PK: NoStaff

Member(NoStaffMem, noKP, noPF, name, status, houseAddress, housePostcode, houseState, gender, religion, race, email, position, positionGrade, officeAddress, officePostcode, officeState, noFax, telNo, housePhoneNum, monthlySalary, bankName, accountNo, joinDate, modalShare, modalFee, fixedSaving, charityFund, memberSaving)
PK: NoStaffMem

	BoardOfDirector(NoStaffBod) PK: NoStaffBod Loan(loan_id, loan_name, loan_rate, loan_max, loan_min, loan_modal_share) PK: loan_id LoanApplication(loan_application_id, loan_type, loan_amount, loan_duration, monthly_payment, loan_employer_verification, loan_application_status, loan_application_date, NoStaffBod, NoStaffMem) PK: loan_application_id FK: loan_type references Loan(loan_id) NoStaffBod references BoardOfDirector(NoStaffBod) NoStaffMem references Member(NoStaffMem) (1:1) Notification(notification_id, message, notification_date, loanApplicationId (FK)) PK: notification_id FK: loanApplicationId references LoanApplication(loan_application_id). Guarantor(g_ic, g_name, g_noPF, g_signature, loanApplicationID) PK: g_ic FK: loanApplicationID references LoanApplication(loan_application_id)
2.5	**Normalization**
	The relation is already in 3NF User(NoStaff , password, usertype) PK: NoStaff Member(NoStaffMem, noKP, noPF, name, status, houseAddress, housePostcode, houseState, gender, religion, race, email, position, positionGrade, officeAddress, officePostcode, officeState, noFax, telNo, housePhoneNum, monthlySalary, bankName, accountNo, joinDate, modalShare, modalFee, fixedSaving, charityFund, memberSaving) PK: NoStaffMem BoardOfDirector(NoStaffBod) PK: NoStaffBod Loan(loan_id, loan_name, loan_rate, loan_max, loan_min, loan_modal_share) PK: loan_id

	LoanApplication(loan_application_id, loan_type, loan_amount, loan_duration, monthly_payment, loan_employer_verification, loan_application_status, loan_application_date, NoStaffBod, NoStaffMem) PK: loan_application_id FK: loan_type references Loan(loan_id) NoStaffBod references BoardOfDirector(NoStaffBod) NoStaffMem references Member(NoStaffMem) (1:1) Notification(notification_id, message, notification_date, loanApplicationId (FK)) PK: notification_id FK: loanApplicationId references LoanApplication(loan_application_id). Guarantor(g_ic, g_name, g_noPF, g_signature, loanApplicationID) PK: g_ic FK: loanApplicationID references LoanApplication(loan_application_id)